



SC1006

Computer Organization and Architecture

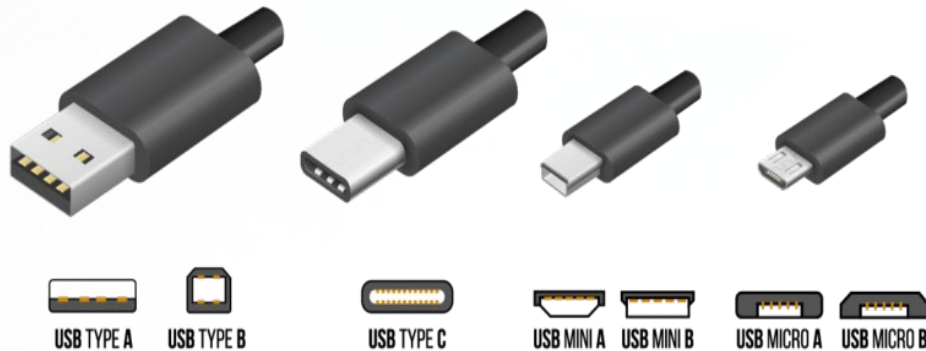
Wk12a1 Review
Comms and UI

Sub-topics in this section

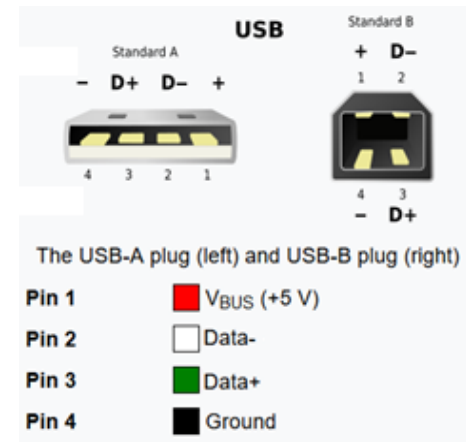
- | | | |
|----------------------------|---|-----------------------|
| 1. USB | } | Comm |
| 2. Wireless (Wifi, BT) | | |
| 3. KeyBoard/Mouse | } | User input |
| 4. Capacitive Touch Screen | | |
| 5. Camera | } | Audiovisual capturing |
| 6. Microphone | | |
| 7. LCD | } | Audiovisual output |
| 8. Audio Speaker | | |

Revision of selected areas in which
students asked questions...

Universal Serial Bus (USB)



Source: <https://www.conwire.com/blog/ultimate-guide-usb-cables/>



Source: <https://electronics.stackexchange.com>

- **Serial Bus** designed to standardize connection of computer peripheral devices (Keyboard, Mouse, Printers, Disk Drives, Network Adapters, Digital Cameras etc).
- Effectively **replaced many interface buses** e.g. Serial Bus (COM Port), Parallel Port etc
- USB1.0 standard only supports a transfer rate of 12Mbps but has progressed over the years and the latest **USB3.2** standard supports up to **20Gbps** transfer.
- **4 basic pinout**: **VBUS** (+5V Power supply), **Data+/Data-** (Data pins) and **Ground** pin.



Which pin does USB Host/Hub supply power to the USB devices?

- A. Data+
- B. Data-
- ✓ C. VBUS
- D. Ground

The very basic USB interface consist of 4 pins: VBUS, Ground, Data+ and Data-.

Data+ and Data- is used for the actual data transmission. As the signal name implies, the USB interface uses differential signalling.

USB devices can be power by the USB connection, and this is done by the Host/Hub supplying a +5V via the VBUS pin to the USB devices.

Why is it that many wireless standards chose to use 2.4GHz as its operating frequency?

- A. It was used in the first version of Wifi and developers continued to use that since the characteristics of that frequency range has been well researched and understood.
- ✓ B. The frequency range is free for use world wide.
- C. Transmission range of this frequency is one of the best among the various frequency range available for use by consumers.
- D. It's the resonance frequency of water, which is available in abundance in nature.

Essentially, 2.4Ghz is used in many wireless appliance because it belongs to a group of RF bandwidth known as the International Scientific Medical (ISM) band. And usage of this RF band is free, i.e. no licensing fee has to be paid.

This compares with other RF bandwidth such the 3G, 4G, 5G cellular band which the telcos such as Singtel, Starhub, M1, TPG etc typically paid huge licensing fees to the authority in order to be able to operate in that band.

Factors affecting transmission range



- Transmission power
 - Transmission range increase as transmission power increase.
- Transmission frequency
 - Higher frequency signal experience higher attenuation when propagating through the air or other medium.
 - All things equal, higher frequency signal has lower range than lower frequency signal.
- Interference
 - Many commonly adopted standards such as Wifi and Bluetooth work in the same 2.4Ghz ISM band. Their transmission will interfere with each other.
 - The closer the transmitter is to each other, the stronger the interference.
 - Even the micro oven in your kitchen operates in the 2.4Ghz range.



Which of the following factor does not affect wireless transmission range?

- A. Transmission Power
- B. Transmission Frequency
- C. Interference from other wireless sources
- ✓ D. Number of repetitions in transmission

[A] Larger transmission power => longer range
[B] Higher frequency => higher attenuation => shorter range
[C] Increase in interference => short range
[D] Number of repetition in transmission may increase the reliability of transmission but will not increase the transmission range

Camera Sub-Modules

- Lens
 - Optical lens to focus the real world objects onto the image sensor.
- Image Sensors
 - Mostly CMOS based these days.
 - transduce photons (light information) to electrical signals (analog).
 - Analog-to-Digital conversion of the analog electrical signal to its digital equivalent for processing in the ISP.
- Image Signal Processor (ISP)
 - Processor designed to specifically handle image processing of collected image data from the sensor.
 - Processing done include Auto Focus, Auto Exposure, Auto White Balance, image format conversion, image post-processing/compression etc.
 - Common image formats are YUV (luminance and chrominance) and RGB (Red, Green, Blue).
- Interface I/O
 - Re-formatting of image data from the ISP for delivery to the host processor.



Which of the following is not a function of the Camera Module in a phone?

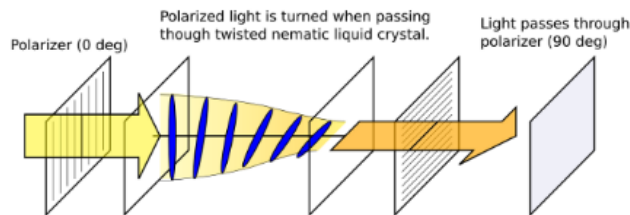
- A. Focus on object
- B. Interfacing to Host Processor
- C. Video Compression
- D. Conditioning of captured image
- ✓ E. none of the above

Why do we need a video compression module in the camera in a phone?



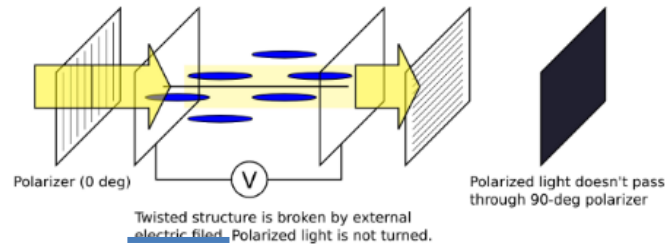
Liquid Crystal Display (LCD) Basics

Polarised light twisted (Light Passes)



Source: awawa.hariko.com

Polarised light not twisted (Light Blocked)



- **Liquid crystal** is an organic substance that has both a liquid form and a crystal molecular structure; The molecules are **able to be controlled to change to a particular orientation/direction**.
- An **electric field V** can be used to **control the molecules orientation**, according to pixel value (brightness) of an image.
 - ❑ Depending on their orientation, these molecules are able to **twist the light passing through them**.
 - ❑ The **amount of light** that is able to pass through the polariser **depends on its orientation with respect to the polariser**.
- This result in **light passing, partial passing or being blocked** from user point of view.



Which of the following is the operating principle of an LCD?

- A. Amount of light reflection off a reflective surface
- B. Angle of light refraction when it pass through a transparent medium
- ✓ C. Amount of polarised light passing through the polariser
- D. Amount of light absorbed by the translucent medium

Liquid crystal molecule is able to orientate light passing through it to create polarised light in a certain orientation. The polarised light is then passed through a polariser to control the amount of light emitted to the external world. By controlling the amount of light passing though different colour filter, various colours can be generated.

Thank you!